

Experimental design: variability, replication and power

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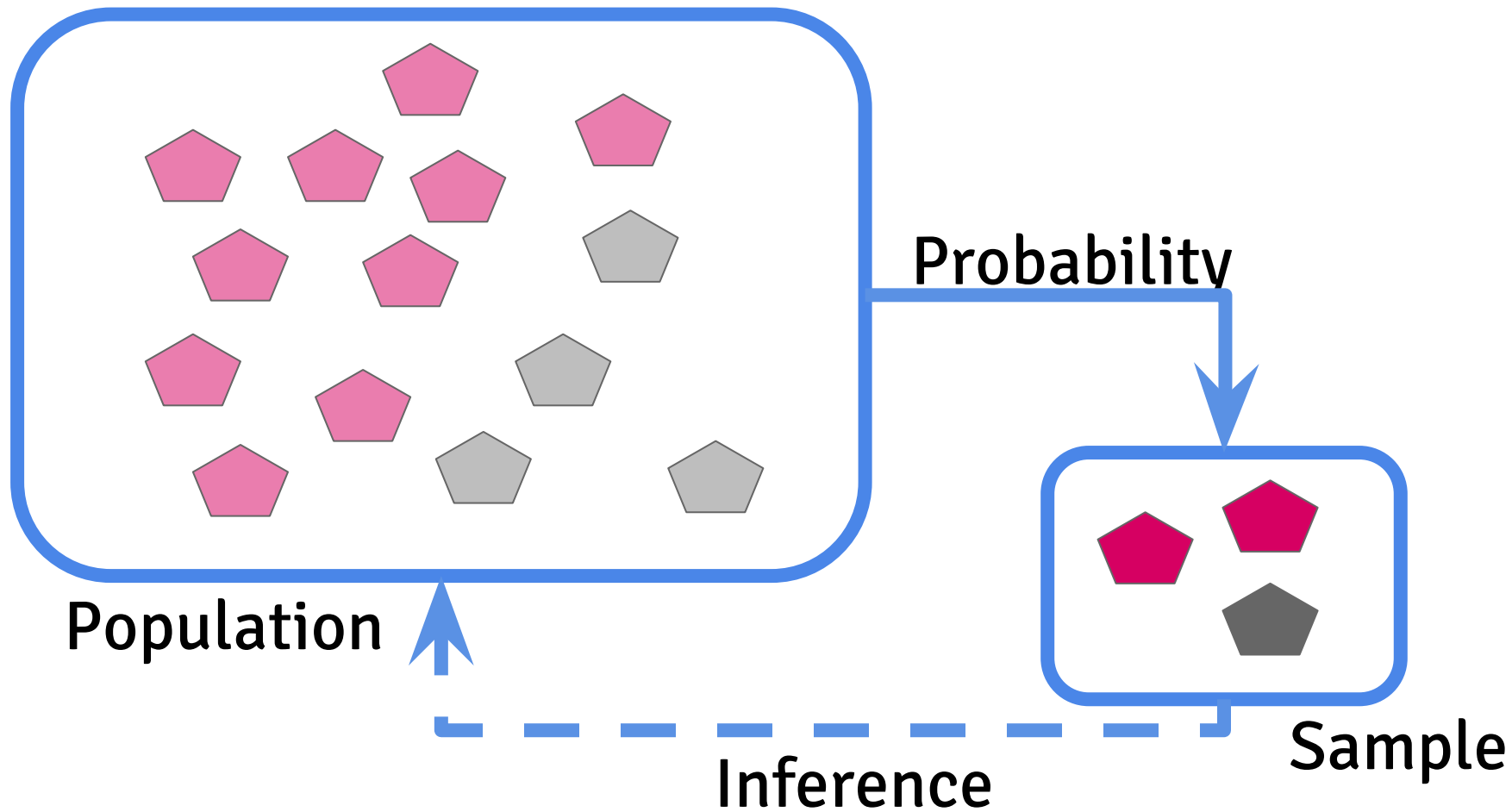
Key ideas

Sources of variation

Power

Replication

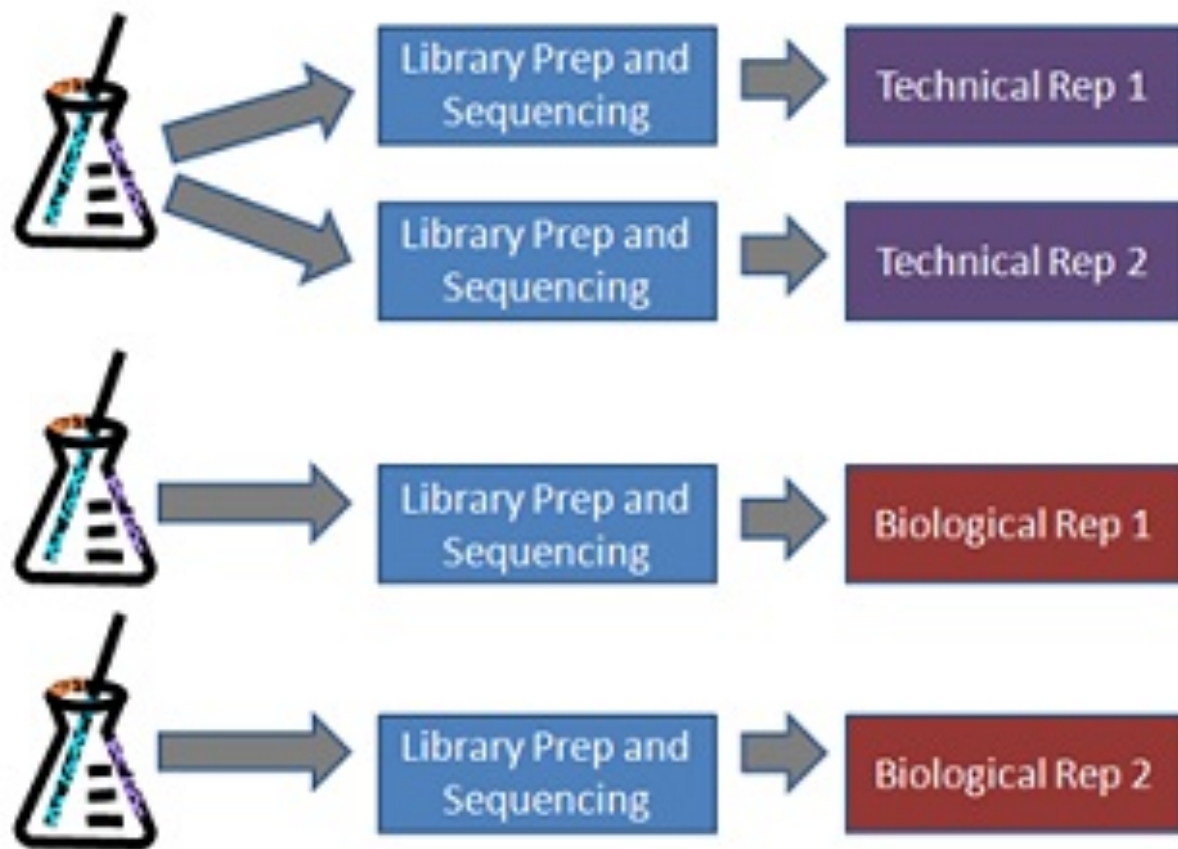
Central dogma of statistics



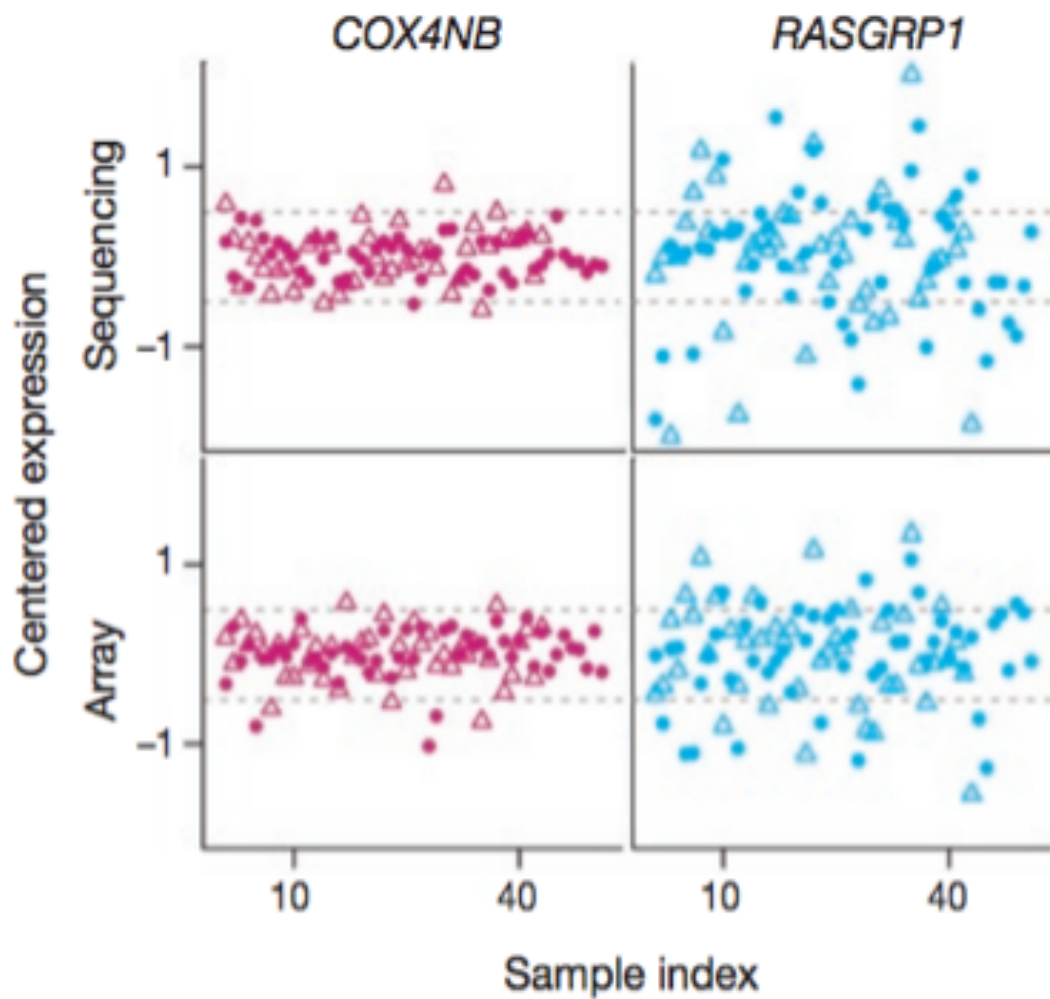
Three types of variability

$$\begin{aligned} \text{Var}(\text{Genomic Measurement}) \\ &= \text{Phenotypic variability} \\ &+ \text{Measurement error} \\ &+ \text{Natural biological variation} \end{aligned}$$

Replicates
Biological
Technological



**New technology doesn't eliminate
variability**



Sample size

N = Number of
Measurements

$$N = \frac{(\$ \text{ you have})}{(\$ / \text{ measurement})}$$

Sample size depends on measurement

Rare mendelian disease

$N \approx 3-5$

RNA-Sequencing study

$N \approx 10 - 1,000$

DNA methylation study

$N \approx 10 - 1,000$

Common disease genome-wide association

$N \approx 10,000 - 1,000,000+$

**New technology doesn't eliminate
variability**



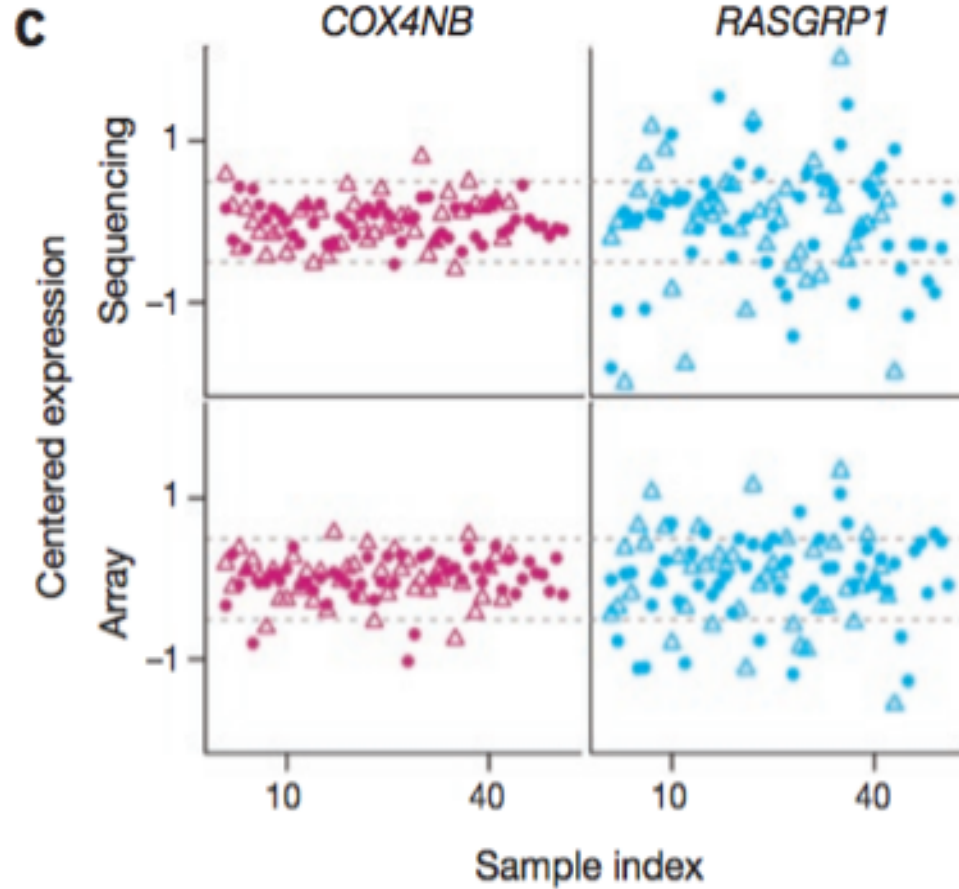
Sequencing technology does not eliminate biological variability

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This is often ignored

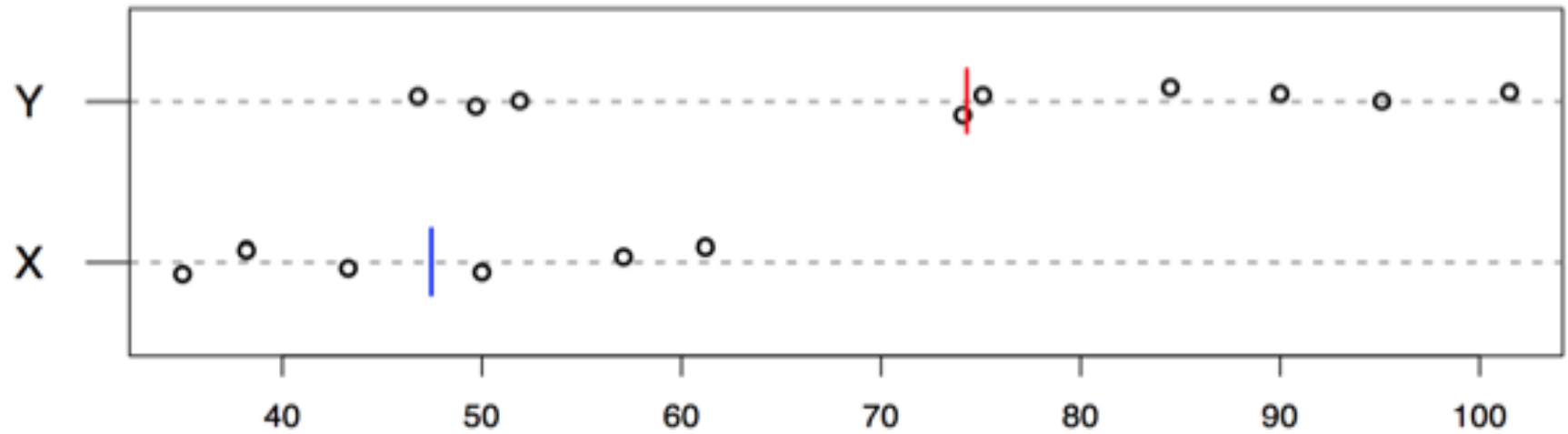
Pubmed ID	Journal (Year)	# of Biological Groups	# of Technical Replicates	# of Biological Replicates
18451266 ¹⁰	Science (2008)	1	2	2
19056941 ¹¹	Science (2008)	1	1	2
18516045 ⁶	Nature Methods (2008)	3	2	1
18599741 ¹²	Science (2008)	2	2	1
18516046 ¹³	Nature Methods (2008)	2	3	1
18978772 ¹⁴	Nature (2008)	15	1	1 (6 in 1 group)
19234113 ¹⁵	PNAS (2009)	2	1	2
19581875 ¹⁶	Nature Biotechnology (2009)	1	3	1
19349980 ¹⁷	Nature Methods (2009)	4	1	2
20436464 ¹⁸	Nature Biotechnology (2010)	4	1	1
20810668 ¹⁹	Genome Research (2010)	9	1	1
20980679 ²⁰	Blood (2010)	2	1	2
21057496 ²¹	Nature Methods (2010)	2	1	1
20452967 ²²	Genome Research (2010)	4	1	2
20363980 ²³	Genome Research (2010)	1	1	1
20220758 ¹	Nature (2010)	1	2	69
20220756 ²	Nature (2010)	1	1	60

Variability and power

Power =

Probability of discovering a real signal if it is there

- Power is typically set at 80%
- Calculations are based on made up assumptions
- Higher power is better
- Low powered studies don't replicate



- ▶ $n = 10$ for each group; effect = $\Delta = 5$; pop'n SD = $\sigma = 10$
`power.t.test(n=10, delta=5, sd=10)`
→ 18%
- ▶ power = 80%; effect = $\Delta = 5$; pop'n SD = $\sigma = 10$
`power.t.test(delta=5, sd=10, power=0.8)`
→ $n = 63.8$ → 64 for each group
- ▶ power = 80%; effect = $\Delta = 5$; pop'n SD = $\sigma = 10$; one-sided
`power.t.test(delta=5, sd=10, power=0.8,
 alternative="one.sided")`
→ $n = 50.2$ → 51 for each group

Power curves

